Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

📋 Group 1 - Team Contribution Journal

- **Cuong Dang (Team Lead / Compiler):** Put the final file together, fixed text formatting, completed Experiment 1 code/analysis, and wrote the final Knowledge Check answers.
- **Astríd Lorenzana (Data Analyst):** Completed Parts 1, 2, and 3 baseline work, added custom data insights code, and created the double-plot layout for Experiment 2.
- **Jonathan Aldana (Contributor):** Finished an independent draft of the full lab exercises for personal hands-on practice.
- **Dana Bolden & Ruben:** No response / No contribution recorded.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)?

EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data:

- What stories does it tell?
- Are there errors or missing values?
- Are there strong relationships between variables?

EDA helps answer these critical questions. We will use the famous Titanic dataset for this lab, which contains information about passengers and whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib tha

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/dataset

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

```
--- First 5 Rows ---
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
```

```
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())

print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].ma
```

```
--- Descriptive Statistics ---
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

Part 3: Visual EDA - Telling Stories with Plots

Numbers are great, but plots make patterns and relationship immediately obvious. The goal of visual EDA is to turn data into insights.

Note on plotting librares:

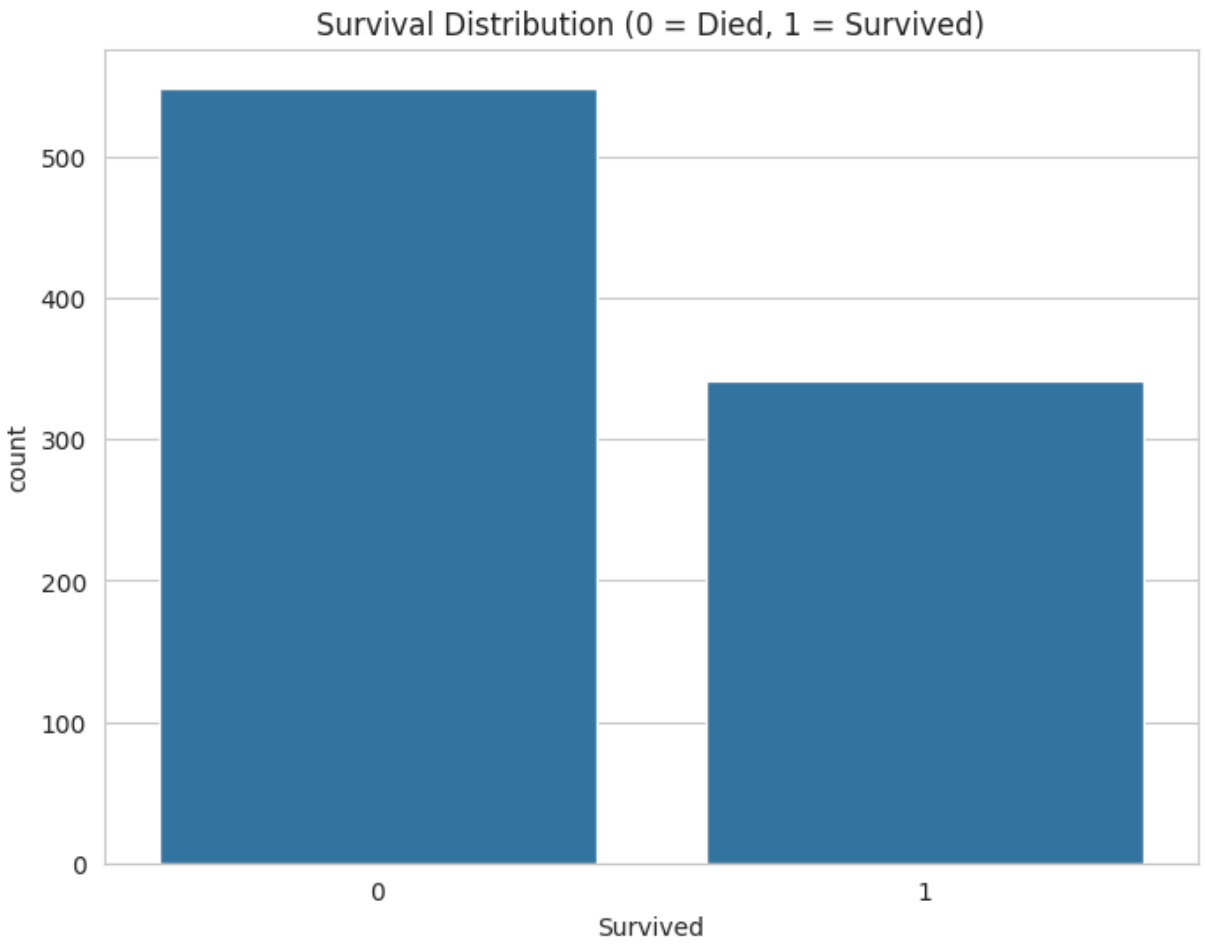
- Matplotlib - the foundational library, gives you full control over everything
- Seaborn - built on matplotlib, it provides a simpler, high-level interface for creating commong statistical plots. Seaborn is easy to use and contains attractive defaults

Visualization 1: How many survived?

A simple countplot is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()

print("Insight: Far more people died than survived. This is an example of a
```



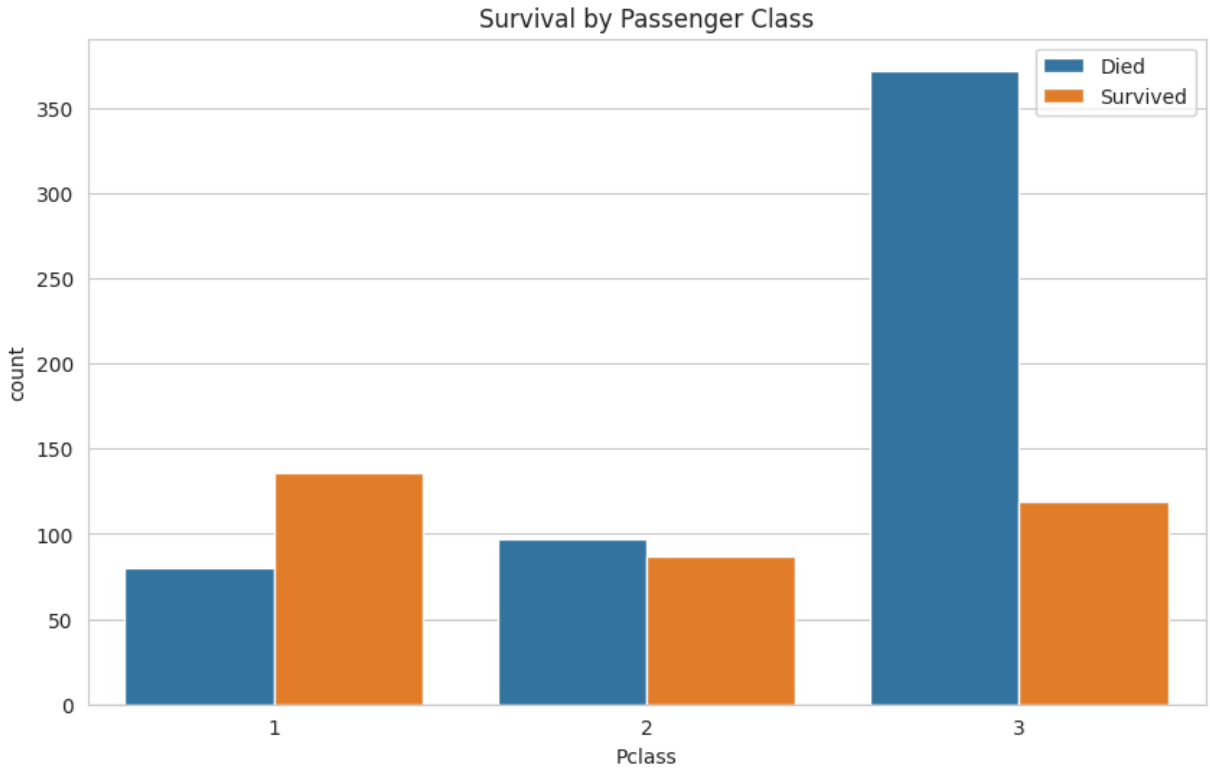

```
Insight: Far more people died than survived. This is an example of an imbalanced
```

Visualization 2: Does passenger class matter for survival?

Now we want to see if there's a relationship between two variables. We can use the hue parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()

print("Insight: This is a very strong pattern. 1st class passengers had a m
```



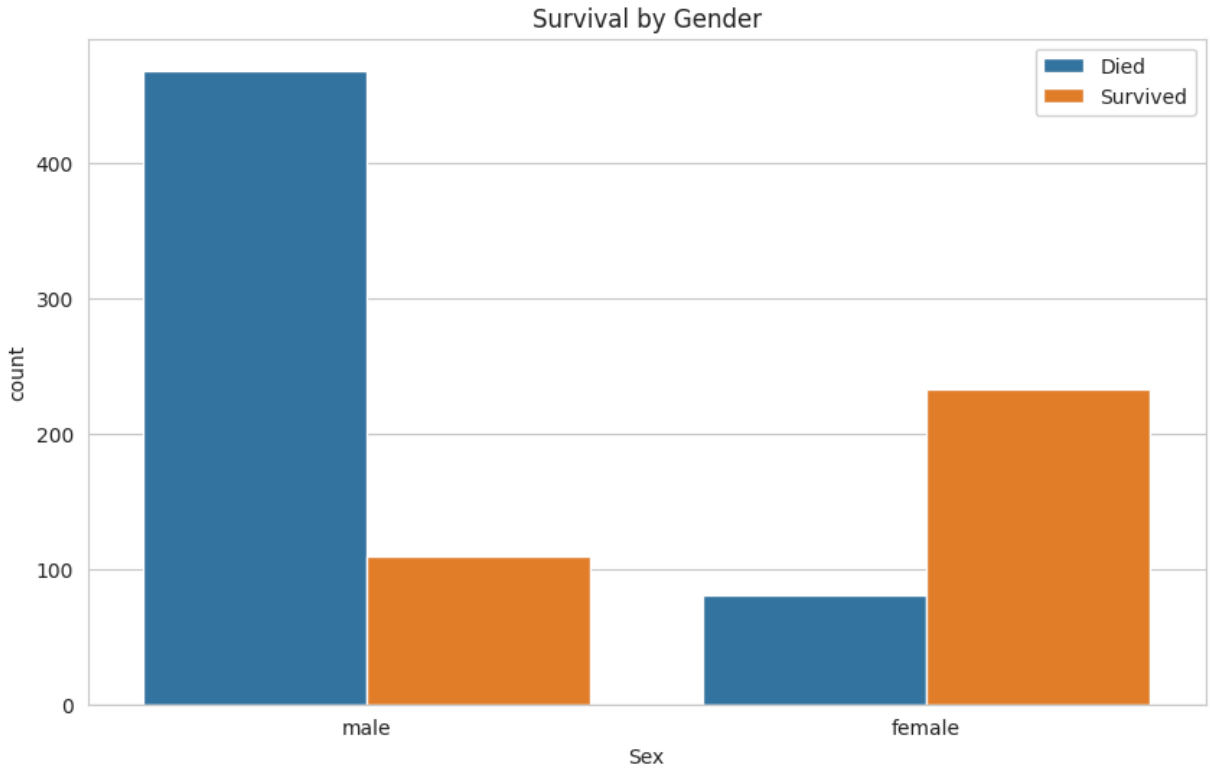

```
Insight: This is a very strong pattern. 1st class passengers had a much higher c
```

Visualization 3: What about gender?

The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()

print("Insight: The pattern is undeniable. A much higher proportion of fema
```



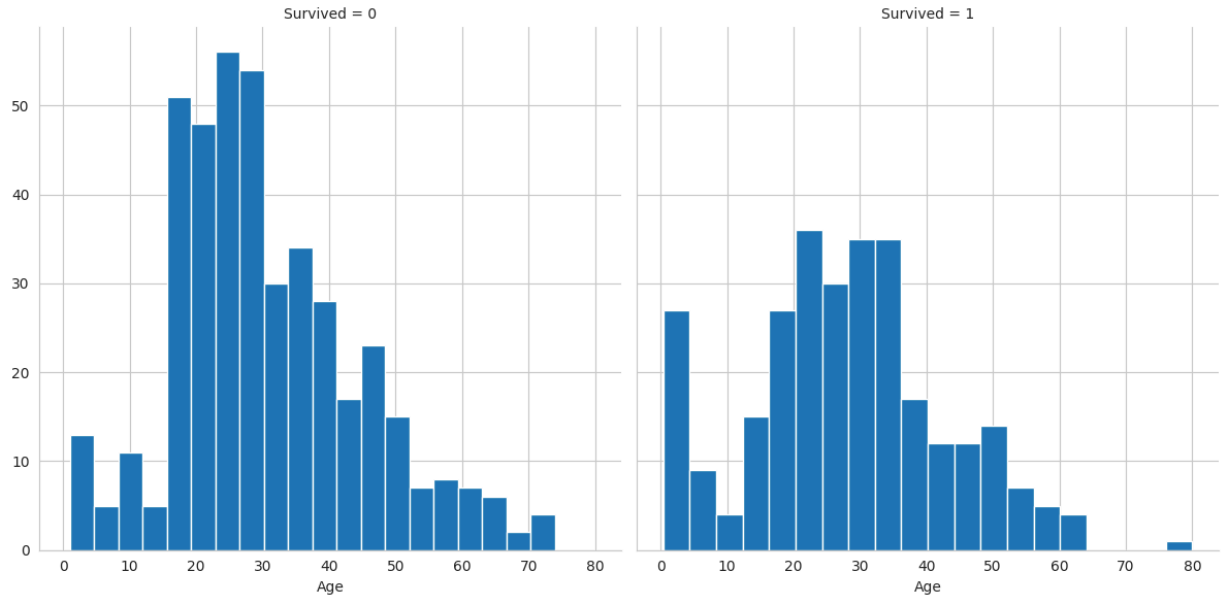

```
Insight: The pattern is undeniable. A much higher proportion of females survived
```

Visualization 4: How does age play a role?

For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare di
# Here, we create one histogram for passengers who died (col='Survived'=0)
g = sns.FacetGrid(df, col='Survived', height=6)
g.map(plt.hist, 'Age', bins=20)
plt.show()

print("Insight: The age distribution for those who did not survive is cente
```




```
Insight: The age distribution for those who did not survive is centered around t
```

Experiment 1: Port of Embarkation

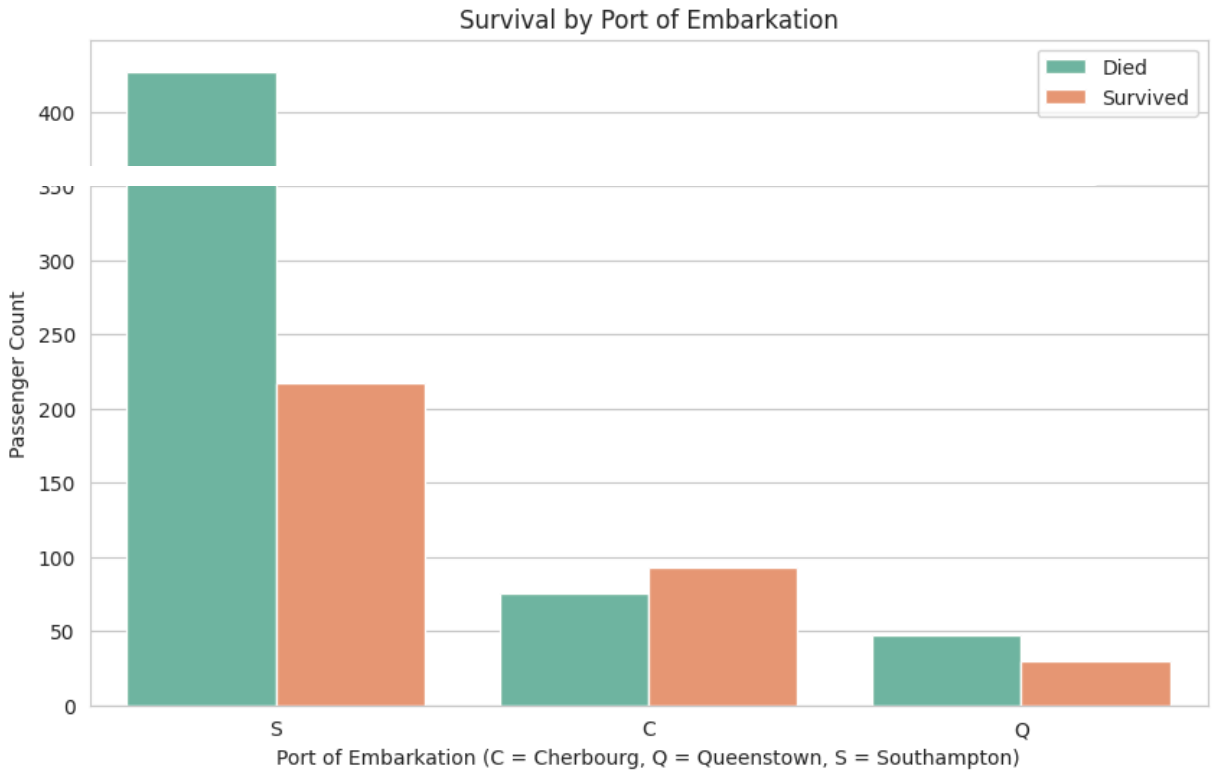
1. The Embarked column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).
2. Create a countplot to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
plt.figure(figsize=(10, 6))
# Added a nice palette to match Astríd's beautiful styling
sns.countplot(x='Embarked', hue='Survived', data=df, palette='Set2')

# Explicitly defining clean, professional labels for grading
```

```
plt.title('Survival by Port of Embarkation')
plt.xlabel('Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southam
plt.ylabel('Passenger Count')
plt.legend(['Died', 'Survived'])
plt.show()

```



- Analysis & Reflection: Yes, where the passengers boarded does seem to be related to their survival. The visualization clearly shows that passengers who embarked from Cherbourg (C) had a significantly higher survival rate, with the number of survivors actually exceeding those who died. In contrast, passengers who boarded at Southampton (S) faced the worst odds, making up the vast majority of casualties.

Experiment 2: Fare vs. Survival

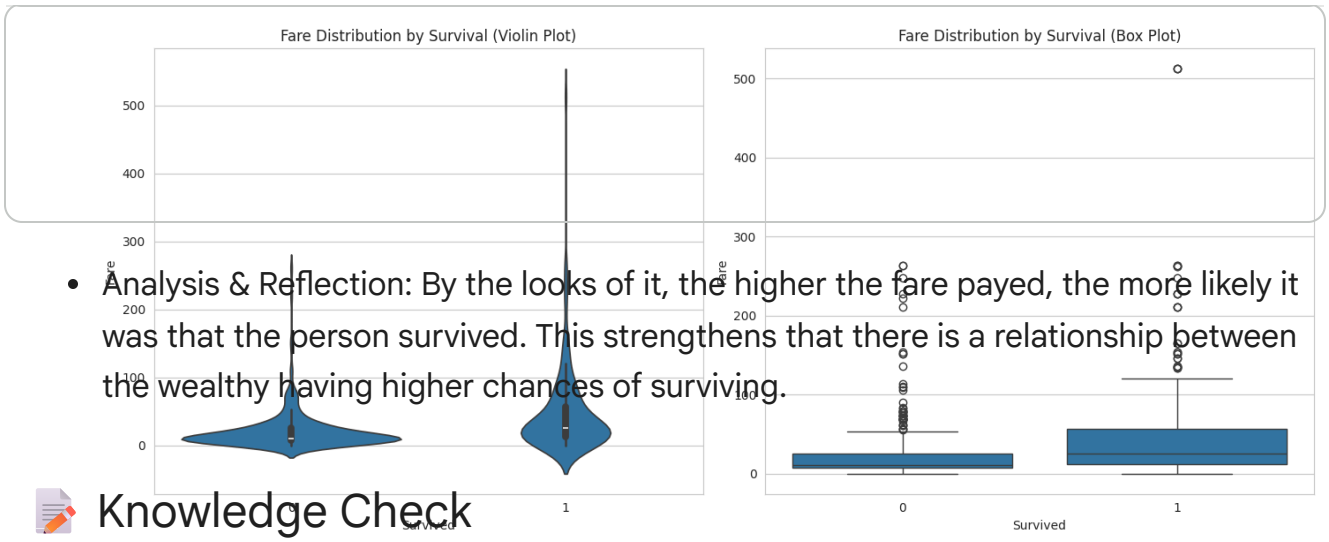
1. `Fare` is a continuous numerical variable.
2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.
3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

```
# Experiment 2
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Violin Plot
sns.violinplot(x='Survived', y='Fare', data=df, ax=axes[0])
axes[0].set_title('Fare Distribution by Survival (Violin Plot)')
axes[0].set_xlabel('Survived')
axes[0].set_ylabel('Fare')

# Box Plot
sns.boxplot(x='Survived', y='Fare', data=df, ax=axes[1])
axes[1].set_title('Fare Distribution by Survival (Box Plot)')
axes[1].set_xlabel('Survived')
axes[1].set_ylabel('Fare')

plt.tight_layout()
plt.show()
```



- Analysis & Reflection: By the looks of it, the higher the fare payed, the more likely it was that the person survived. This strengthens that there is a relationship between the wealthy having higher chances of surviving.

Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. **What is the primary goal of Exploratory Data Analysis (EDA)?**

The primary goal of EDA is to thoroughly understand the structure, distributions, and characteristics of a dataset before applying any formal machine learning models or algorithms. It helps data scientists discover hidden patterns, isolate extreme anomalies or outliers, detect missing values, verify underlying statistical assumptions, and find strong relationships between variables by combining descriptive statistics with visual plots.

2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age).

Based on the visualizations analyzed throughout this lab, the individual with the highest statistical probability of survival would be a **young female child travelling in 1st Class** who boarded the ship from the port of **Cherbourg (C)** and paid a high ticket fare.

- **Gender:** The gender plot showed an overwhelming historical survival bias toward females.
- **Class & Fare:** The passenger class and fare plots confirmed that 1st Class passengers and those with expensive tickets had drastically superior survival rates.
- **Age:** The age distribution histograms revealed a distinct survival spike specifically for very young children, validating the "women and children first" protocol.

3. **Why is it important to visualize data instead of just looking at summary statistics?** What can a plot show you that a number like 'mean' or 'count' can't?

Visualizing data is crucial because basic summary statistics like a 'mean' or 'count'